Let there be...hacks!!!

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CSAW'22 MENA Hack My Robot

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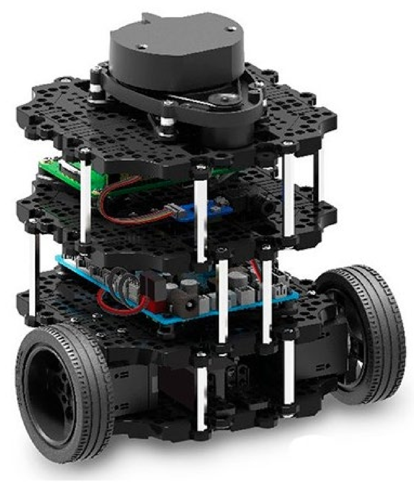
SENTRY SECURE NEXT GENERATION RESILIENT SYSTEMS LAB

Introduction

- In recent years, robots are ubiquitous in construction, industrial systems, automation, health systems, etc.
- Robotic agents are a crucial part of mission-critical cyber-physical systems
- **Turtlebot 3** is a Raspberry Pi-based robot that uses the ROS platform for the coordination of its movement and other auxiliary tasks
- The interoperable robotic architectures increase the attack surface (e.g., sensors, actuators, applications)
- Our team demonstrates how different attacks types can compromise **Turtlebot's** operation

Raspberry Pi ROBOTIS TURTLEBOT3 ROS Robot Operating System



Compromising Availability

ARP* Poisoning Attack

- Send forged ARP packets
- Deceive communication endpoints
- Become Man-in-the-Middle
- Intercept, forward, and modify certain packets/commands
- Flood the communication channel between robot and user/server

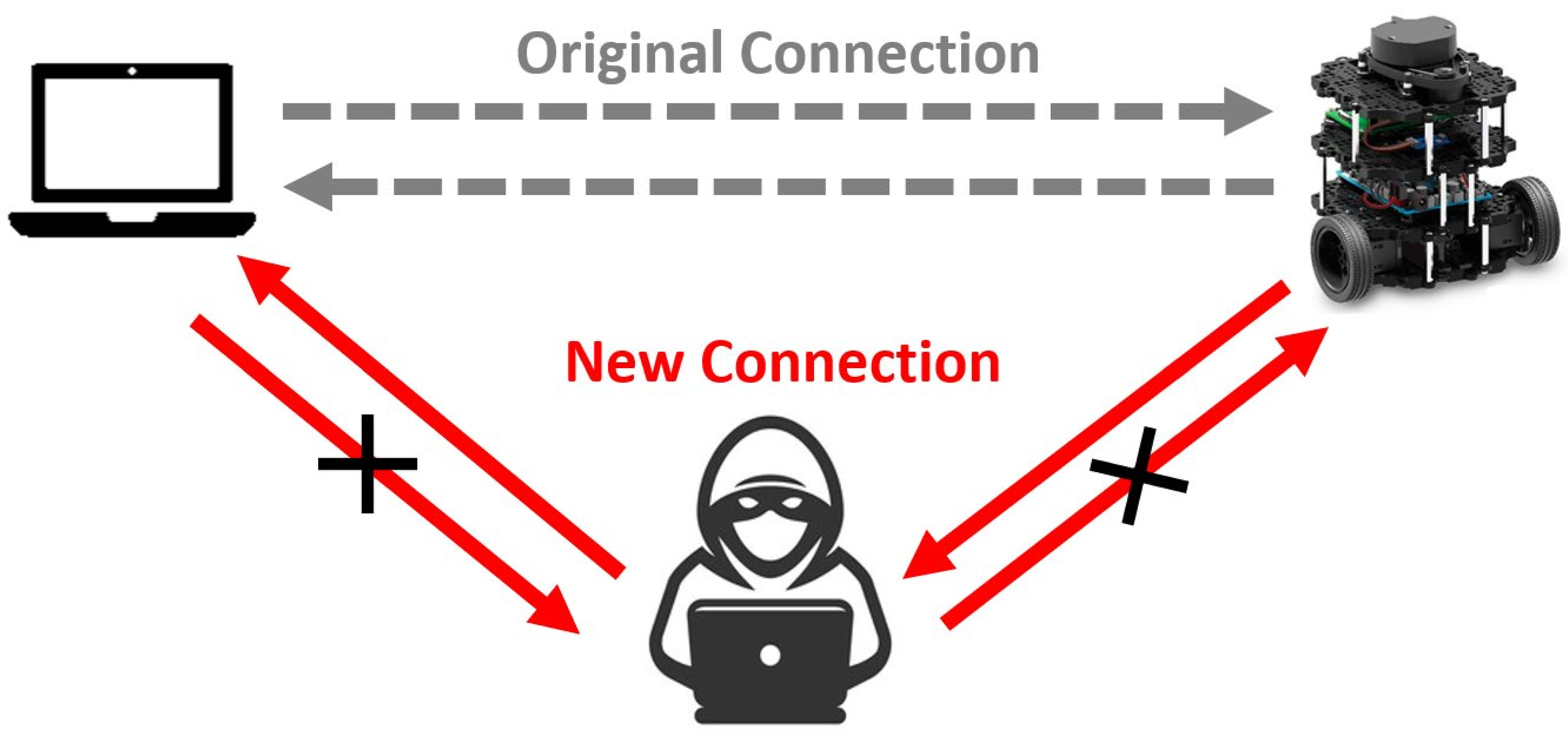
Tools: Kali Linux, airmon-ng

De-authentication Attack

- Flood communication channel
- Force the de-authentication of the host (robot)
- Capture control packets attempting the re-authentication
- Inspect packets and identify critical sections (e.g., wi-fi passwords)

Tools: Kali Linux, airmon-ng

Adversarial access: Remote – access to network
Impact: Denial-of-Service



*ARP = address resolution protocol

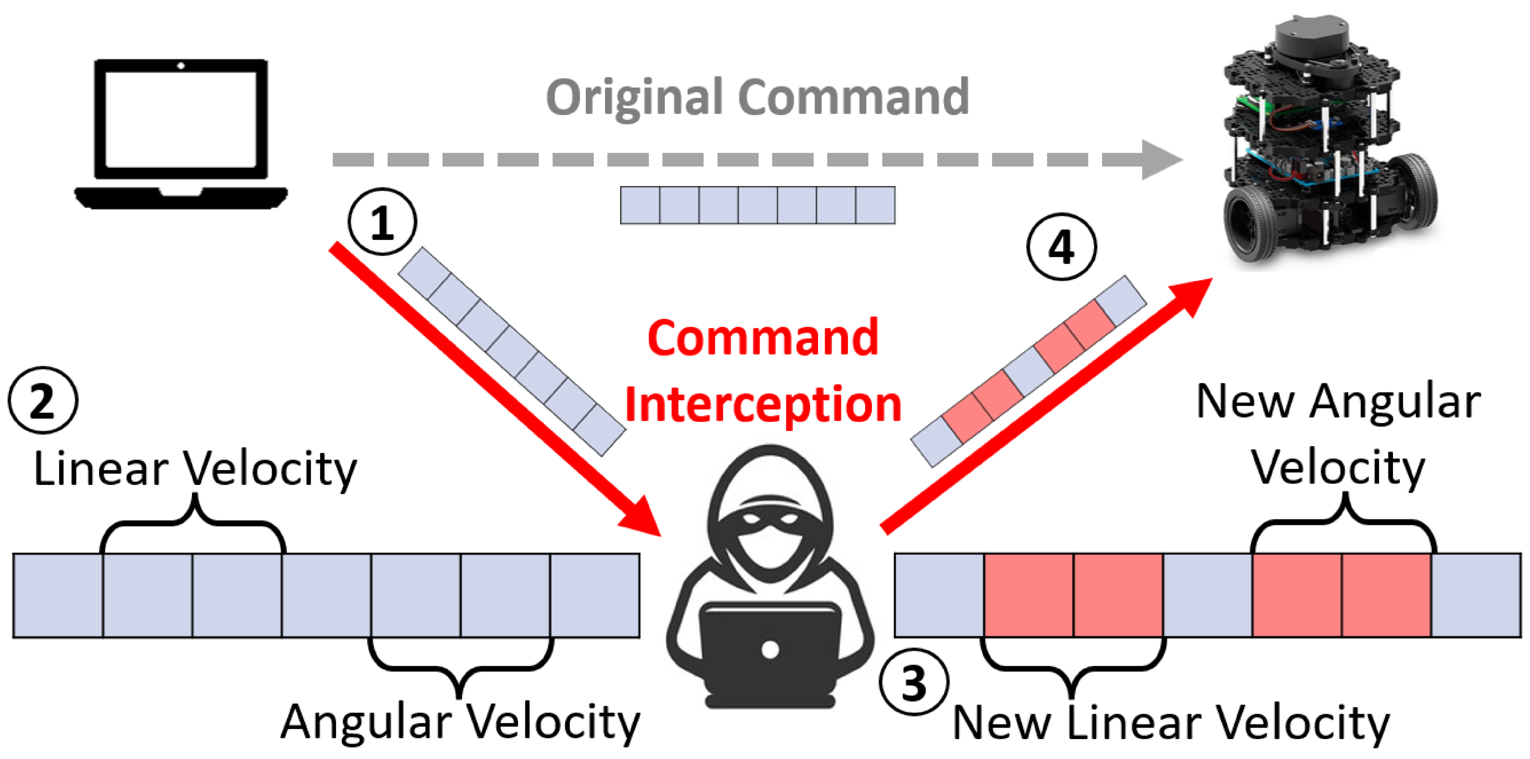
Taking Control

Packet Modification Attack

- Successful ARP spoofing attack
- Capture control packets
- Modify packet contents, i.e., robot linear/angular velocity
- Forward maliciously crafted packets to target

Tools: Arpspoof, Scapy, Wireshark, iptables, nfqueue

Adversarial access: Remote – access to network
Impact: Loss of controllability, Denial-of-Service



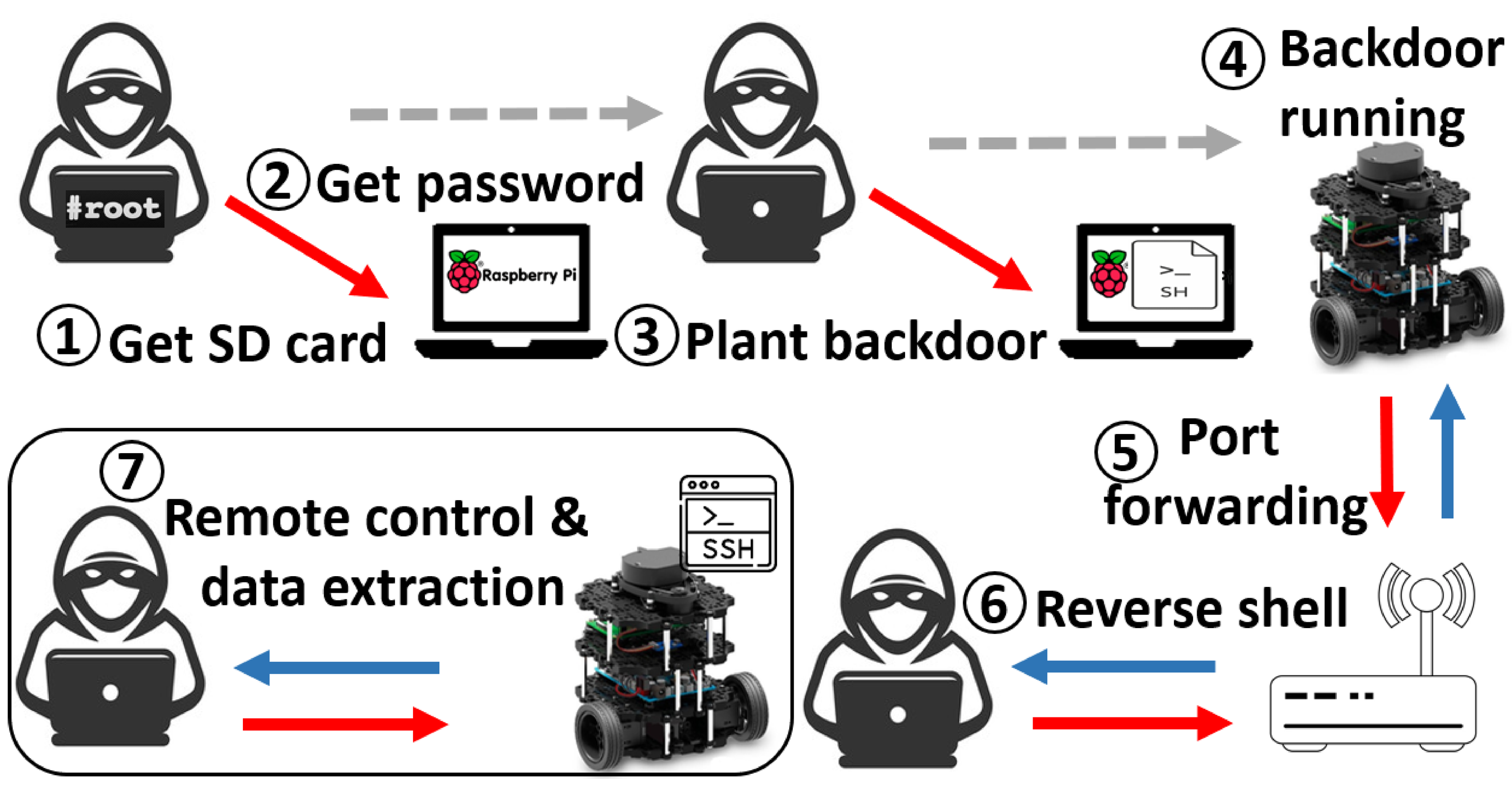
Becoming Stealthy

Backdoor Attack

- Access the SD card and change the boot process
- Attain **root password** and gain root privileges
- Plant backdoor on robot using Metasploit
- Automatically start a **reverse shell** on system reboot
- Attain **persistence** and forward communication to malicious user
- Full remote control of the robotic agent

Tools: Kali Metasploit framework, meterpreter, frp

Adversarial access: One-time physical access
Impact: Remote access and control of the robot



Conclusions

- Vulnerabilities can exist in any of the underlying robotic architectural levels
- We present how robotic missions can be compromised via different threat vectors
- Secure-by-design schemes are paramount for robots in untrustworthy environments

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